Kyle W. **Rader**

Principal Software Engineer · Developer Tooling & AI Systems

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Skills

Languages	Rust, C# (.NET), Python, TypeScript, Bash
AI/Agents	Claude Code, GitHub Copilot CLI, Cursor
Tooling	CLI UX design, software distribution, SDK design, TDD, PR automation, linting pipelines
Infra	Git, Docker, Azure DevOps, Azure Pipelines, GitHub Actions, Scalar, Entra ID

Experience

Principal Software Engineer — Agency CLI (AI Coding Agents) · Microsoft *Remote*

March 2025 - Present · 1 yr

Building the AI engineering platform that brings coding agents to Microsoft's internal developer population at enterprise scale.

- Lead engineer on the **Agency CLI** — the core of Microsoft's internal AI engineering platform, serving 35,000+ engineers across interactive workflows and CI/CD pipelines, enabling AI coding agents (Claude Code, GitHub Copilot) to operate effectively across massive brownfield codebases.
- Drive CLI UX, software distribution, and agent integration design — pioneering a novel company-wide developer tool distribution model for 35,000+ engineers where no internal solution previously existed.
- Partnered with GitHub and Anthropic to iterate rapidly on Copilot CLI and Claude Code integrations; shipped improvements and channeled direct user feedback enabling agents to reliably operate in Microsoft's largest, most complex repositories.
- Built an MCP proxy layer within Agency CLI that natively handles Microsoft Entra ID authentication, solving MCP auth for internal developers and making all internal MCPs work out-of-the-box in Claude Code, GitHub Copilot, and any MCP-compatible agentic client.
- Defined and drove CLI UX standards across the entire Agency platform: config design, command consistency, usability patterns, and type-system-enforced interface contracts to ensure correctness and coherence at scale.

Principal Software Engineer — Azure Hardware & Infrastructure · Microsoft *Remote*

June 2024 - March 2025 · 10 mos

Embedded systems SDK work for next-generation Azure hardware infrastructure.

- Built an async **Rust SDK** for Azure developers to interface with a custom operating system running on new Microsoft-designed data center chips, enabling hardware-level control from idiomatic Rust.

Principal Software Engineer — Office DevPlat · Microsoft *Remote*

December 2022 - June 2024 · 1 yr 7 mos

Designed CLI platforms and tooling abstractions empowering developers building foundational Microsoft products (Word, Excel, PowerPoint).

- Designed and built **autolinter** (`aka.ms/autolinter`) — a generic Rust-based linter toolkit that automatically applies lint fixes to pull requests in Azure DevOps, used across Office engineering.
- Built **azureauth** (`aka.ms/azureauth`) — a general-purpose Microsoft Entra ID authentication CLI used by **10,000+ developers daily**, delivering the most seamless auth experience available internally.
- Created **lasso** (`aka.ms/lasso`) — a C# CLI framework providing logging, UX, and telemetry out of the box; adopted across the team's entire suite of developer tools.
- Championed Rust adoption for developer tooling within Office Engineering: defined best practices, reviewed implementations, and grew the internal Rust practitioner community.

Senior Software Engineer — Office DevPlat · Microsoft *Redmond, WA*

May 2018 - December 2022 · 4 yrs 7 mos

Developer experience engineering for the Office organization's global engineering system.

- Built and operated engineering systems used by thousands of developers worldwide to build and ship Microsoft Office.
- Developed an EditorConfig-based line-ending linter for the 180GB Office monorepo, catching 3–5% of PRs with encoding issues before merge.
- Built Azure Pipelines branch protection policies safeguarding Office's branching schema for 5,000+ engineers, preventing catastrophic pushes before merge.
- Co-founded the Experiences & Devices GLEAM sub-chapter; organized Fix-Hack-Learn engineering events three times per year.

Software Engineer · ActionSprout *Bellingham, WA*

July 2016 - March 2018 · 1 yr 9 mos

Tools and training to help nonprofits exceed their goals on Facebook.

- Architected web micro-services, recommendation systems, and business intelligence pipelines for a nonprofit social media platform.
- Pioneered Docker/Docker-Compose standardization to solve developer environment consistency across the team.
- Built a Facebook Graph API RubyGem, Rails micro-service generator, and JWT authentication standards used daily in production.

Software Development Life Cycle Manager · 8las Inc *Bellingham, WA*

July 2015 - July 2016 · 1 yr 1 mo

Augmented reality startup.

- Established agile processes, codebase organization, and version control standards to streamline development velocity for an early-stage AR startup.
- Ran augmented reality demos and pitched the product to investors in Seattle and California.

Production Engineer · WompMobile *Bellingham, WA*

March 2015 - August 2015 · 6 mos

Mobile web platform converting desktop sites to mobile.

- Front-end developer converting desktop websites to mobile-optimized experiences using HTML, CSS, and JavaScript.

Web Developer · Logos Bible Software *Bellingham, WA*

June 2013 - January 2014 · 8 mos

Bible study software and publishing company.

- Developed internal web services tools on the DevOps team; built an open-source generic tile-based web page using Ruby on Rails.

Projects

Kitchen Table Games · Founder & Lead Engineer *https://kitchentable.games*

2026 - Present

Real-time multiplayer puzzle games platform built under Automata Software LLC.

- Building a privacy-first, real-time multiplayer games platform under Automata Software LLC — no ads, no data selling, no third-party tracking.
- Full-stack Rust application with real-time game state sync supporting original real-time multiplayer word and number based puzzle games for 1–20 players per table.

The Great Puzzle Hunt · Lead Engineer & Co-Founder *https://greatpuzzlehunt.com*

2015 - 2018

Annual puzzle hunt at Western Washington University drawing ~500 participants competing on a real-time game platform.

- Full-stack web platform handling registration, team formation, payment webhooks, real-time scoring, volunteer QR scanning, and administration — serving ~500 participants annually.
- Built in React/Node.js, continuously iterated over 10+ years of annual events.

Education

Western Washington University · B.S. Computer Science *Bellingham, WA*

2010 - 2014

- Computer Science Dept. Citizenship Award & Scholarship
- Technology Leader of Tomorrow — Technology Alliance Group NW (TAGNW)
- Robotics Club President, IEEE Vice President, ACM Activities Coordinator & Mentor

Western Washington University · M.S. Computer Science (Coursework completed) *Bellingham, WA*

2014 - 2018

- Completed all required coursework; left program before thesis defense for health reasons.
- Founded *Out in Science* — a WWU graduate community for LGBTQ+ students in STEM.